

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	2024	Date:	20/07/2026

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A video editing application is exporting a 4K movie, where the CPU continuously fetches video frames and audio data from an SSD, applies filters and compression, and writes the processed output to a display preview and a storage device. Construct a concept map linking the various units of the computer, using labelled arrows to illustrate the flow of instructions and data. How do the five functional units coordinate to retrieve multimedia data, process video and audio operations, and produce the final output during video rendering?
2. A software engineer notices that a data analytics application is taking longer than expected to process large datasets. To identify the cause, the engineer analyses how each instruction progresses through the Fetch, Decode, and Execute stages and examines performance factors such as CPI, Clock Cycles, Clock Rate, and CPU Execution Time. Construct a concept map linking the Instruction Execution Cycle with these performance parameters. How do the Fetch–Decode–Execute stages influence CPU execution time, and how can the performance factors be used to evaluate and improve processor performance?
3. A robotics engineer is selecting a microprocessor for an autonomous mobile robot and must compare the 8085 and 8086 architectures to determine which offers better execution speed and memory management. Construct a concept map comparing the 8085 and 8086 by linking their instruction sets, register organization, addressing modes, memory organization, and memory segmentation (8086). Show how these architectural features influence program execution, memory access, multitasking capability, and efficiency in real-time applications.
4. A software developer is optimizing an embedded control program and must choose the most suitable addressing mode for different operations to achieve faster execution and efficient memory usage. Construct a concept map linking Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode influences memory access time, instruction size, execution speed, and processor efficiency, and include real-world applications where each addressing mode is most appropriate.
5. A firmware engineer is debugging an embedded system where machine instructions are stored in memory in binary but displayed in hexadecimal for easier analysis. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU instruction interpretation, data conversion, and debugging. Show how data is converted between binary and hexadecimal, how instructions are stored and executed by the CPU, and why hexadecimal notation is preferred in low-level programming and system design. How do binary and hexadecimal number systems support instruction representation, memory organization, CPU execution, and debugging in computer systems?

Code of Conduct

I R. Gunuprasad/1924720941 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R. Gunuprasad

Function units of Computer during video Rendering

Video Editing Application



Input unit (SSD)

(Reads video frames & audio data)



Memory unit (RAM)

(Stores instructions, frames & audio temporarily)



Control unit

(Fetches → Decodes → Controls execution flow)



ALU

(Filters, Compression, Calculation, Encoding)



Memory unit

(Stores processed multimedia data)



Output unit

(Display preview + storage device)

Flow

SSD → RAM → Control unit → ALU → RAM → Display & storage.

Instruction Execution Cycle & Performance

program Instructions



Fetch (memory)



Decode (CU)



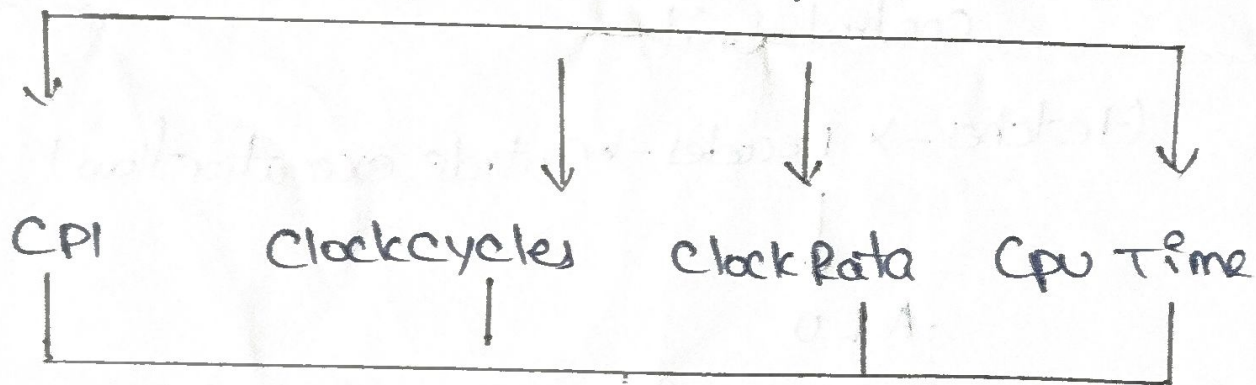
Execute (ALU)



Write Back Result



CPU performance parameters



Processor Performance.

Formula

CPU Execution time

$$= \text{Instruction count} \times \text{CPI} \times \text{Clock cycle time}$$

(or)

$$= \text{Clock cycles} / \text{Clock Rate}$$

8085 vs 8086 Architecture

8085 Microprocessor	Feature	8086 Microprocessor
8-bit	Data bus	16 bit
16-bit	Address Bus	20 bit
64-KB	Addressable Memory	1 MB
6x8-bit (B, C, D, E, H, L)	Registers	8x16bit (AX, BX, CX, DX, SP, BP, SI, DI)
Simple (Non-segmented)	Architecture	Advanced (segmented)
74 Instructions	Instruction set	111 Instructions
Up to 6 MHz	Clock speed	5-10 MHz
Lower	Performance	Higher.

8086 is more powerful and suitable for complex application.

Q4. Addressing Modes

①

Immediate Addressing

Operand is part of the instruction itself

Example: `MVI A, 25H`
Advantage: Fast



②

Direct Addressing

Addressing of operand is given in the instruction

Example: `LDA 2050H`
Advantage: Simple.

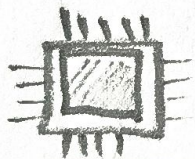


③

Register Addressing

Operand is in a register

Example: `MOV A, B`
Advantage: Very Fast



④

Indirect Addressing

Address of operand is in a register or memory

Example: `MOV A, M`
Advantage: Flexible



⑤

Indexed Addressing

Effective address = Base address + Index register

Example: `MOV A, [SI + 05H]`



Advantage: Efficient for arrays.

Binary & Hexadecimal Number System

Binary Number System
uses 0 and 1
Base = 2 101101
Used inside CPU
Easy for electronic circuit

Binary is Machine
friendly

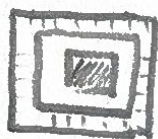
Binary to Hex
Conversion

Hexadecimal
Number system
uses 0-9 & A-F
Base = 16
Compact representation
Easy reading
for humans
2A3F

Hex is human
friendly.

Instruction Representation

Programs and data are stored in memory
using binary or hexadecimal



CPU Execution

Fetch instruction from memory
Decode and execute
Perform operation using ALU

Application.



Debugging and programming
Memory addressing
System analysis

